



Day 36 – Binary Tree Basics

♦ 1. Binary Tree kya hota hai?

Binary Tree ek hierarchical data structure hai jisme:

Har node ke max 2 children hote hain:

Left Child

Right Child

👉 Example:

```

      10
     / \
    5  20
   / \
  2  8
  
```

♦ 2. Basic Terminology

Term	Meaning
Root	Sabse top node (10)
Parent	Jiska child ho
Child	Jo node kisi ke niche ho
Leaf Node	Jiske koi children nahi (2, 8, 20)
Height	Tree ki max depth
Depth	Root se distance

♦ 3. Types of Binary Tree (Important)

✓ 1. Full Binary Tree

Har node ke 0 ya 2 children

✓ 2. Complete Binary Tree

Last level ko chhod ke sab full

Last level left se fill hota hai

✓ 3. Perfect Binary Tree

Sab levels completely filled

◆ 4. Binary Tree Node (Java Code)

```
class Node {
    int data;
    Node left, right;

    Node(int data) {
        this.data = data;
        left = right = null;
    }
}
```

◆ 5. Tree Creation Example

```
public class BinaryTree {
    Node root;

    public static void main(String[] args) {
        BinaryTree tree = new BinaryTree();

        tree.root = new Node(10);
        tree.root.left = new Node(5);
        tree.root.right = new Node(20);
        tree.root.left.left = new Node(2);
        tree.root.left.right = new Node(8);

        System.out.println("Tree Created Successfully");
    }
}
```

◆ 6. Tree Traversals (MOST IMPORTANT 🔥)

```
Inorder (LNR)
Left → Root → Right
void inorder(Node root) {
    if (root == null) return;

    inorder(root.left);
    System.out.print(root.data + " ");
    inorder(root.right);
}
```

2. Preorder (NLR)

👉 Root → Left → Right

```
void preorder(Node root) {
    if (root == null) return;

    System.out.print(root.data + " ");
    preorder(root.left);
    preorder(root.right);
}
```

3. Postorder (LRN)

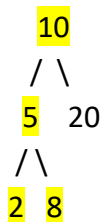
👉 Left → Right → Root

```
void postorder(Node root) {
    if (root == null) return;

    postorder(root.left);
    postorder(root.right);
    System.out.print(root.data + " ");
}
```

◆ 7. Output Example

For this tree:



Traversal	Output
Inorder	2 5 8 10 20
Preorder	10 5 2 8 20
Postorder	2 8 5 20 10

◆ 8. Important Interview Points ⚡

- ✓ Tree = Recursive structure → recursion zaruri
- ✓ Traversal = base of all tree problems
- ✓ Height, diameter, LCA → next level topics